

1. A rectangular swimming pool 50m long, 20m wide, and 5m deep is filled with water to a depth of 4m. Set up a definite integral to compute the work required to pump the water out over the top of the swimming pool.

The density of the water is 1000 kg/m^3 , and the gravitational acceleration is 9.8 m/s^2 .

Fill in the blanks in the following answer line. **Do not evaluate the definite integral.**

Answer: Work = $\int_{\text{---}}^{\text{---}}$ _____ d _____ J.

Formula : $W = \int_0^b \rho g A(y)(h - y) dy$

2. Find the general solution to $\frac{dy}{dx} = 10x^4y^2$.

Write your final answer in the form of $y = f(x)$.